



Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments, and product preferences from **3,900 transactions** to guide strategic business decisions.

Dataset at a Glance

3,900

Transactions

Total purchase records analyzed

18

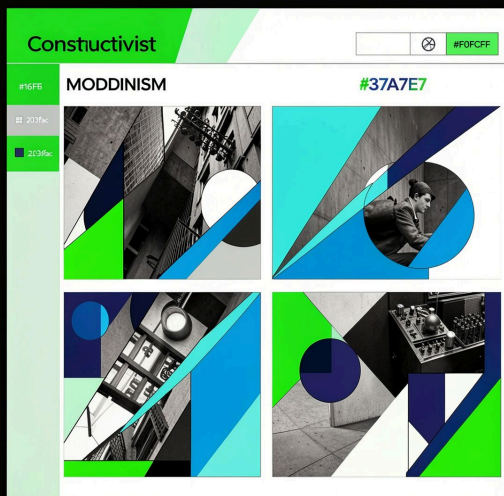
Columns

Features spanning demographics, purchases, and behavior

37

Missing Values

In Review Rating column — imputed by category median



Key Features

- **Demographics:** Age, Gender, Location, Subscription Status
- **Purchase Details:** Item, Category, Amount, Season, Size, Color
- **Behavior:** Discount Applied, Previous Purchases, Frequency, Review Rating, Shipping Type

Data Preparation in Python

01

Load & Explore

Imported dataset via pandas; used `df.info()` and `.describe()` for structure and summary stats.

02

Clean & Standardize

Imputed missing Review Ratings using category medians; renamed columns to snake_case.

03

Feature Engineering

Created `age_group` by binning ages; derived `purchase_frequency_days` from purchase data.

04

Database Integration

Connected to PostgreSQL and loaded the cleaned DataFrame for SQL analysis.



Revenue & Spending Insights



Revenue by Gender

Compared total revenue generated by male vs. female customers to identify spending differences across genders.

High-Spending Discount Users

Identified customers who used discounts but still spent **above the average purchase amount** — valuable targets for retention.

Shipping Type Comparison

Compared average purchase amounts between **Standard** and **Express** shipping to understand spend behavior by shipping preference.

Product & Discount Analysis

Top 5 Products by Rating

Found products with the highest average review ratings to highlight top-rated items in marketing campaigns.

Discount-Dependent Products

Identified the **5 products** with the highest percentage of discounted purchases — key for reviewing discount policy and margin control.

Top 3 Products per Category

Listed the most purchased products within each category for targeted product positioning.





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Subscribers vs. Non-Subscribers

Compared average spend and total revenue across subscription status to measure the value of subscriber engagement.

Subscribers

Analyzed average spend and total revenue contribution from subscribed customers.

Non-Subscribers

Compared spend patterns against non-subscribed customers to identify conversion opportunities.

Customer Segmentation



Three Customer Tiers

Classified customers into **New**, **Returning**, and **Loyal** segments based on purchase history.

Repeat Buyers & Subscriptions

Checked whether customers with **more than 5 purchases** are more likely to subscribe – a key signal for loyalty program design.

Revenue by Age Group



Age Group Contribution

Calculated total revenue contribution of each age group to identify which demographics drive the most spend.

Targeting Opportunity

High-revenue age groups represent the strongest opportunity for **targeted marketing campaigns** and personalized offers.

Interactive Power BI Dashboard

Built an interactive dashboard in Power BI to present all insights visually – enabling stakeholders to explore revenue trends, customer segments, product performance, and demographic breakdowns at a glance.



Business Recommendations



Boost Subscriptions

Promote exclusive benefits to convert high-frequency buyers into subscribers.



Loyalty Programs

Reward repeat buyers to move them into the "Loyal" segment.



Review Discount Policy

Balance sales boosts with margin control on discount-dependent products.



Targeted Marketing

Focus on high-revenue age groups and express-shipping users.



Product Positioning

Highlight top-rated and best-selling products in campaigns.